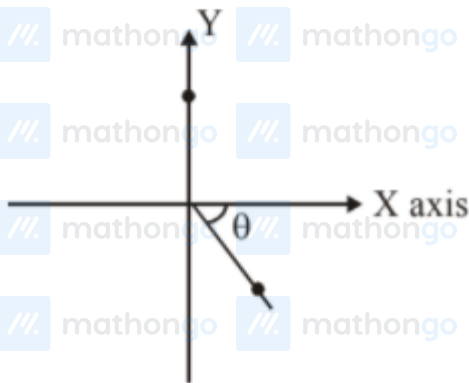


Q1: 16 March (Shift 1) - Numerical

A ball of mass 10 kg moving with a velocity $10\sqrt{3} \text{ ms}^{-1}$ along X-axis, hits another ball of mass 20 kg which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y-axis at a speed of 10 m/s. The second piece starts moving at a speed of 20 m/s at an angle θ (degree) with respect to the X-axis.

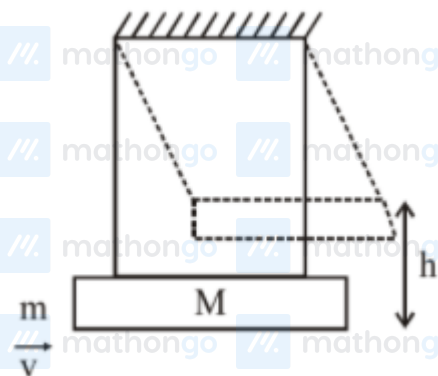
The configuration of pieces after collision is shown in the figure. The value of θ to the nearest integer is _____

After Collision



Q2: 16 March (Shift 2) - Single Correct

A large block of wood of mass $M = 5.99 \text{ kg}$ is hanging from two long massless cords. A bullet of mass $m = 10 \text{ g}$ is fired into the block and gets embedded in it. The (block + bullet) then swing upwards, their centre of mass rising a vertical distance $h = 9.8 \text{ cm}$ before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is : (Take $g = 9.8 \text{ ms}^{-2}$)



(1) 841.4 m/s

Questions with Answer Keys

MathonGo

(2) 811.4 m/s

(3) 831.4 m/s

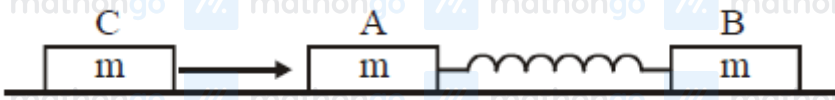
(4) 821.4 m/s

Q3: 17 March (Shift 2) - Single Correct

Two identical blocks A and B each of mass m

resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant

K . A third block C of mass m moving with a speed v along the line joining A and B collides with A . The maximum compression in the spring is



(1) $\sqrt{\frac{M}{2K}}$

(2) $\sqrt{\frac{mv}{2K}}$

(3) $\sqrt{\frac{mv}{K}}$

(4) $\sqrt{\frac{m}{2K}}$

Q4: 17 March (Shift 2) - Numerical

The disc of mass M with uniform surface mass

density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at

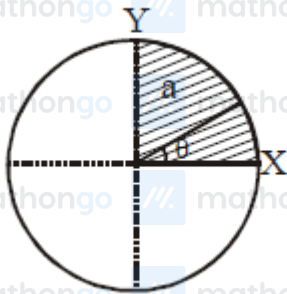
the position $\frac{x}{3} \frac{a}{\pi}, \frac{x}{3} \frac{a}{\pi}$ where x is _____

(Round off to the Nearest Integer)

Questions with Answer Keys

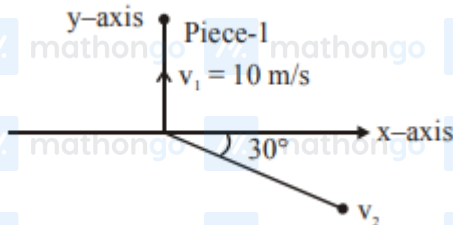
MathonGo

[a is an area as shown in the figure]



Q5: 18 March (Shift 1) - Numerical

A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m/s along the x -axis, hits another ball of mass 20 kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along y -axis with a speed of 10 m/s. The second piece starts moving at an angle of 30° with respect to the x -axis. The velocity of the ball moving at 30° with x -axis is x m/s. The configuration of pieces after collision is shown in the figure below. The value of x to the nearest integer is _____



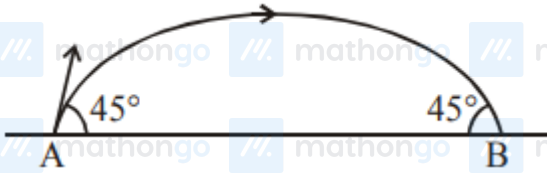
Q6: 18 March (Shift 2) - Single Correct

An object of mass m_1 collides with another object of mass m_2 , which is at rest. After the collision the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is :

- (1) 3: 1
- (2) 2: 1
- (3) 1: 2
- (4) 1: 1

Q7: 18 March (Shift 2) - Numerical

The projectile motion of a particle of mass 5 g is shown in the figure.



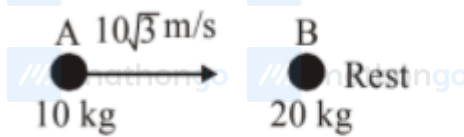
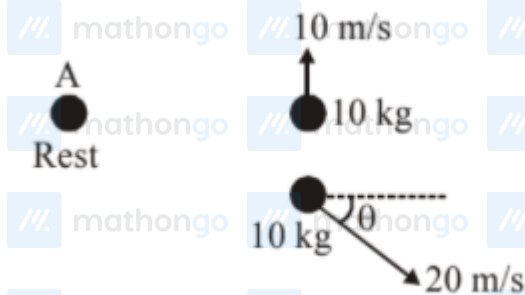
The initial velocity of the particle is

$5\sqrt{2} \text{ ms}^{-1}$ and the air resistance is assumed to be negligible. The magnitude of the change in momentum

between the points A and B is $x \times 10^{-2} \text{ kgms}^{-1}$. The value of x, to the nearest integer, is ____.

Answer Key**Q1 (30)****Q2 (3)****Q3 (1)****Q4 (4)****Q5 (20)****Q6 (1)****Q7 (5)**

Q1 (30)

Before Collision**After Collision**

From conservation of momentum along x axis; $\vec{P}_i = \vec{P}_f$

$$10 \times 10\sqrt{3} = 200 \cos \theta$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

Q2 (3)

From energy conservation,

[after bullet gets embedded till the system comes momentarily at rest]

$$(M + m)gh = \frac{1}{2}(M + m)v_1^2$$

[v_1 is velocity after collision]

$$\therefore v_1 = \sqrt{2gh}$$

Applying momentum conservation, (just before and just after collision)

$$mv = (M + m)v_1$$

$$v = \left(\frac{M+m}{m} \right) v_1 = \frac{6}{10 \times 10^{-3}} \times \sqrt{2 \times 9.8 \times 9.8 \times 10^{-2}}$$

$$\approx 831.55 \text{ m/s}$$

Hints and Solutions

MathonGo

Q3 (1)

C comes to rest

$$V_{cm} \text{ of } A \& B = \frac{v}{2}$$

$$\Rightarrow \frac{1}{2} m v_{ret}^2 = \frac{1}{2} k x^2$$

$$x = \sqrt{\frac{\mu \times v^2}{k}} = \sqrt{\frac{m}{2k}} v$$

Q4 (4)

C.O.M of quarter disc is at $\frac{4a}{3\pi}, \frac{4a}{3\pi}$

$$= 4$$

Q5 (20)

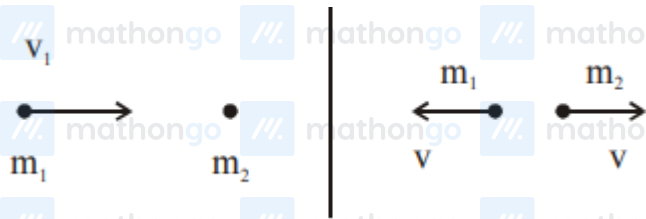
Let velocity of 2nd fragment is $\vec{v}$ then by
conservation of linear momentum

$$10(10\sqrt{3})\hat{i} = (10)(10\hat{j}) + 10\vec{v}$$

$$\Rightarrow \vec{v} = 10\sqrt{3}\hat{i} - 10\hat{j}$$

$$|\vec{v}| = \sqrt{300 + 100} = \sqrt{400} = 20 \text{ m/s}$$

Q6 (1)



$$m_1 v_1 = -m_1 v + m_2 v$$

$$v_1 = -v + \frac{m_2}{m_1} v$$

$$\frac{(v_1 + v)}{v} = \frac{m_2}{m_1}$$

$$e = \frac{2v}{v_1} = 1$$

$$v = \frac{v_1}{2}$$

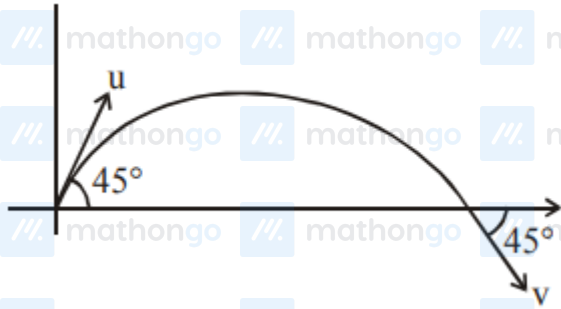
Hints and Solutions

MathonGo

$$\frac{v_1 + v_1/2}{v_1/2} = \frac{m_2}{m_1}$$

$$3 = \frac{m_2}{m_1}$$

Q7 (5)



$$|\vec{u}| = |\vec{v}| \dots (1)$$

$$\vec{u} = u \cos 45^\circ \hat{i} + u \sin 45^\circ \hat{j} \dots (2)$$

$$\vec{v} = v \cos 45^\circ \hat{i} - v \sin 45^\circ \hat{j} \dots (3)$$

$$|\Delta \vec{P}| = |m(\vec{v} - \vec{u})| \dots (4)$$

$$\Delta P = 2mu \sin 45^\circ$$

$$= 2 \times 5 \times 10^{-3} \times 5\sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$= 50 \times 10^{-3}$$

$$= 5 \times 10^{-2}$$